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SciCapenter: Supporting Caption Composition for Scientific Figures with Machine-Generated Captions and Ratings

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ABSTRACT

Crafting effective captions for figures is important. Readers heavily depend on these captions to grasp the figure's message. However, despite a well-developed set of AI technologies for figures and captions, these have rarely been tested for usefulness in aiding caption writing. This paper introduces SciCapenter, an interactive system that puts together cutting-edge AI technologies for scientific figure captions to aid caption composition. SciCapenter generates a variety of captions for each figure in a scholarly article, providing scores and a comprehensive checklist to assess caption quality across multiple critical aspects, such as helpfulness, OCR mention, key takeaways, and visual properties reference. Users can directly edit captions in SciCapenter, resubmit for revised evaluations, and iteratively refine them. A user study with Ph.D. students indicates that SciCapenter significantly lowers the cognitive load of caption writing. Participants' feedback further offers valuable design insights for future systems aiming to enhance caption writing.

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1 INTRODUCTION

Captions for scientific figures, such as charts, bar charts, and pie charts, are crucial to readers. Readers comprehend and recall the underlying information significantly better when reading charts with captions, as compared with reading charts alone [2, 13, 30, 40]. Unfortunately, authors do not pay needed attention to crafting figure captions in their papers. A recent study showed that over half of the figure captions in arXiv papers were rated as not helpful by Ph.D. students [19]. Meanwhile, with advances in deep learning, computational models now have a rich set of capabilities surrounding scientific figures: generating captions of decent quality [24, 37, 49], analyzing the information in figure images [6, 26, 35], and evaluating the usefulness of figure captions [18]. However, literature has little to say about how these technologies could be used to help academics who *write* figure captions for their papers. The user studies of these technologies were almost all conducted from a *reader* perspective, *i.e.*, having participants read the machine-generated outputs and judge their quality [19, 51, 52]. The user needs of writers and readers are known to be different [3, 8, 14, 44]; a piece of machine-generated text might be of lower readability but could serve as a useful draft for writers to work off of [9, 10, 28]. A few prior works, such as Intentable [5], InkSight [33], and AutoTitle [34], built systems that generate captions considering writers' intentions and allow authors to refine visualization captions and titles, but they did not focus on composing figure captions for scholarly articles.

This paper introduces SciCapenter, a system that puts together a set of cutting-edge AI technologies surrounding scientific figures to help users write figure captions in scholarly articles (Figure 1). Users first upload the PDF file of their working draft to SciCapenter, which will automatically extract all the figures and their current captions. Users can then browse and pinpoint figures they wish to modify. Upon selecting a figure, SciCapenter displays the figure image, its current caption, paragraphs in the article referencing the exact figure (*e.g.*, "Figure 3 demonstrates..."), and a series of AI-generated insights. Specifically:

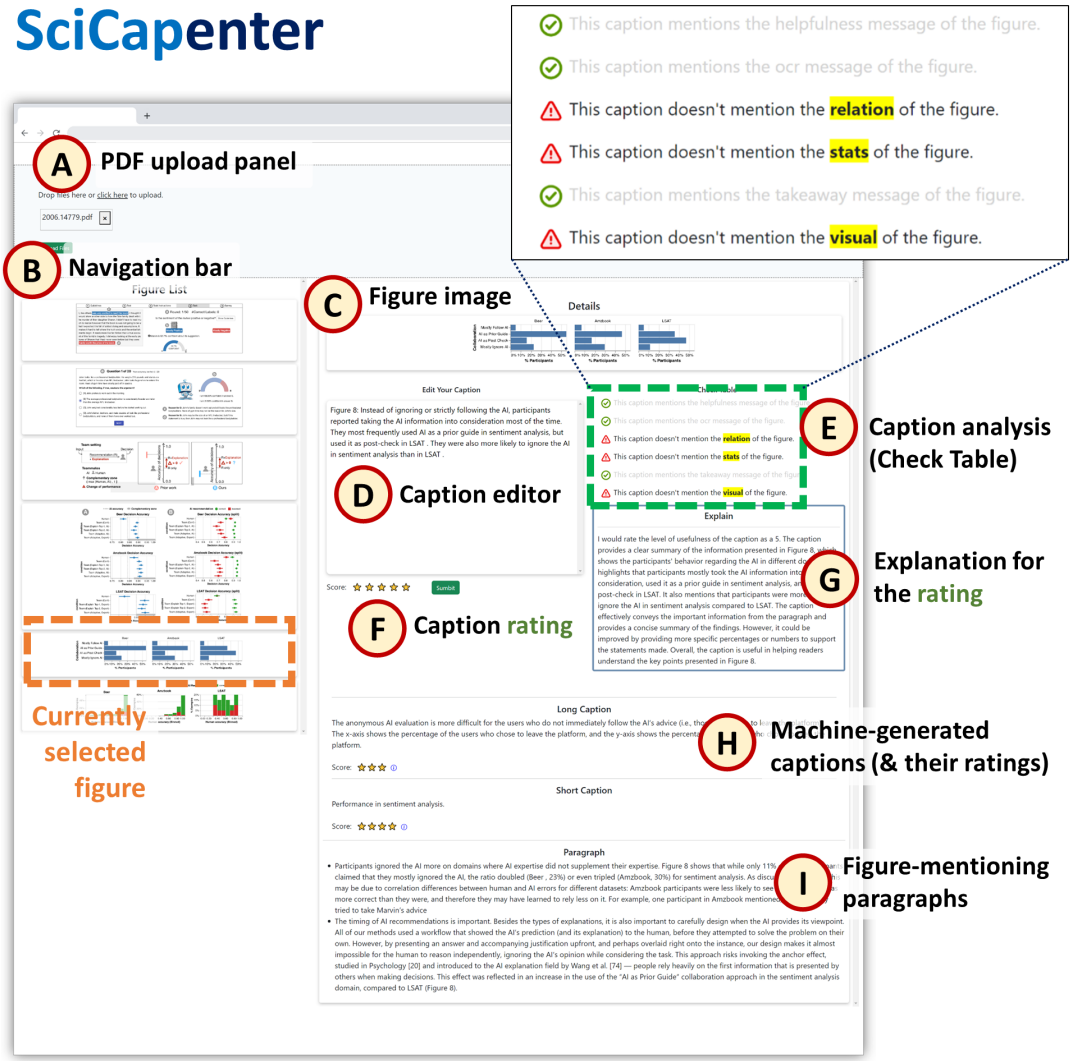


Figure 1: Overview of SciCapenter system interface. PDF Upload Panel (A): A drag-and-drop interface for uploading PDF files. Navigation Bar (B): A horizontal bar showing a list of figures extracted from the uploaded document. Figure Image (C): The main area displaying the image of the selected figure. Caption Editor (D): A text box for editing the caption of the selected figure. Caption Rating (F): A feedback system that allows GPT to rate the quality of the caption, represented by a star rating. Caption Analysis (Check Table) (E): Icons indicating the presence or absence of key elements in the caption, such as helpfulness or takeaway message. Explanation for the Rating (G): A textual explanation providing insight into why a particular star rating was given to the caption. Machine-generated Captions & Their Ratings (H): This section includes long and short captions generated by AI models, each accompanied by their respective star ratings. Figure-mentioning Paragraphs (I): Paragraphs in the document that mention the target figure, providing context or additional information.

- SciCapenter offers two **machine-generated captions** (Figure 1.H): one long, another short, using a cutting-edge AI model. We decided to provide captions with two different lengths to address the dual needs of caption writers: longer captions are more informative for readers [19], while shorter captions help authors meet space constraints in papers.
- SciCapenter provides a **checklist** (Figure 1.E), predicted by an AI model, detailing essential caption features (*e.g.*, does the caption highlight the central message? Does it reference figure text?).
- SciCapenter **rates** all captions (Figure 1.F), including the author's original and all AI-generated ones, and also shows the **rationale behind each rating** (Figure 1.G). Users can modify captions directly in SciCapenter (Figure 1.D), submit them for updated ratings, and refine them iteratively.

A user study involving 15 STEM Ph.D. students spanning Computer Science to Mechanical Engineering indicated that SciCapenter significantly reduced the cognitive load of caption writing, measured by the NASA task load index (NASA TLX). The majority expressed a desire for similar tools in their future works.

The contribution of this work is three-fold. First, our work demonstrates that even if machine-generated text is not superior to human-written text, it could still aid writing. Second, the user study results highlight the advantages of presenting automated predictions, such as target text's characteristics, when providing writing assistance. Finally, participants' feedback offers valuable design insights for future systems aiming to enhance caption writing.

2 BACKGROUND

Figures in documents, such as bar charts, line charts, and pie charts, often contain key information the authors desire to convey. To fully decode the message embedded in figures, reasoning across languages (*e.g.*, document texts, figure captions) and vision (*e.g.*, figure images) is required. This complex task presents a great challenge for the AI community, attracting researchers to develop technologies specifically to address figures. These advancements include AI models that parse information like trends, axes, or statistics in figure images [24, 45], generate a variety of descriptions based on the image of the figure (*i.e.*, vision-to-language models) [31, 35, 37] or the data underlying it (*i.e.*, data-to-text models) [11, 41], answer questions about figure images [22, 23, 38], and suggest appropriate figure types based on the data [20, 25, 46].

Among these efforts, a significant portion was devoted to the study of scientific figures and their captions in scholarly articles. The SciCap dataset [16], introduced in 2021, was the first large-scale collection of real-world figures and captions, containing 2,170,719 figures and captions extracted from 295,028 arXiv papers. Unlike previous works that used synthetic data [4, 21, 22], SciCap facilitated the development of AI models capable of handling real-world scientific figures. This includes models for generating captions based on figure images [15, 16, 32, 36, 47, 51] or paper content [15, 19, 32, 51]. There are also models that evaluate the quality of given captions [18]. In October 2023, the first SciCap Challenge was held [17], where six global teams competed on a performance leaderboard, showcasing the advancements in this field.

From a Human-Computer Interaction (HCI) perspective, these emerging technologies have great potential for aiding caption writing. However, they were rarely studied in a writing context. Instead, the focus of most user studies on these technologies has been from a reader's standpoint. This includes methods like showing participants machine-generated figure captions and asking them to assess their quality [27, 36, 48, 50], presenting various caption variations for participants to compare or rank [27, 48, 50], or testing how much information readers can remember after reading a figure or a caption [27, 48]. Given that the needs of writers and readers are distinct [3, 8, 14, 44], studies focusing on writing scenarios could greatly enhance the application of these technologies in support of caption writing. A few studies and systems have been developed to aid caption writing. For instance, Intentable combines automated caption generation with manual input, empowering authors to guide the captioning process based on their insights and intentions derived from the visualization [5]; InkSight allows users to highlight areas of interest in visualizations through sketching, then automatically generates documentation that users can further revise [33]; and AutoTitle facilitates the interactive creation of titles for visualizations [34]. However, the focus of these systems was not on supporting figure caption composition in scholarly articles.

In this paper, we introduce SciCapenter and delve into exploring this area.

3 SCICAPENTER SYSTEM

SciCapenter is a comprehensive system that includes a web interface and a caption editing function. Figure 1 shows a screenshot of the entire system. To begin using SciCapenter, users start by uploading their PDF file in the PDF upload panel (Figure 1.A). Once the PDF file is processed, the extracted figures are displayed on the navigation bar (Figure 1.B). Users can click on a figure to access the detailed information. Upon clicking, the selected figure (Figure 1.C), along with complete information, appears on the right side of the window. On the right side of the window, users have access to several features that assist them in composing captions, including a checking table (Figure 1.E), GPT evaluation and explanation (Figure 1.F and 1.G), generated captions - long/short versions (Figure 1.H), and (4) referred paragraphs (Figure 1.I). Users are able to edit the caption in the caption editor (Figure 1.D) and resubmit it for re-evaluation.

In the backend, once a user submits a pdf file, SciCapenter will store the file in our MongoDB and extract figures, captions, abstracts, and contents using `pdffigure2` [7].[1] The extracted figure list will then be presented in the navigation bar (Figure 1.B). Regular expressions are then used to identify the corresponding referred paragraphs (Figure 1.I) based on the figure index, similar to how Huang et al. extracted figure-mentioning paragraphs [19].

The check table (Figure 1.E) detects whether the written caption describes or expresses the following six aspects [18]:

(1) **Helpfulness:** Does this caption help you understand the figure?
(2) **OCR (Optical Character Recognition):** Does this caption mention any words or phrases that appear in the figure? (Examples include the figure title, X or Y axis titles, legends, names of models, methods, subjects, etc.)
(3) **Relation:** Does this caption describe a relationship among two or more elements or subjects in the figure? (For example, "A is lower than B," "A is higher than B," or "A is the lowest/highest in the figure.")
(4) **Stats:** Does this caption mention any statistics or numbers from the figure? (For example, "20% of..." or "The value of .. is 0.33...".)
(5) **Takeaway:** Does this caption describe the high-level takeaways, conclusions, or insights the figure tries to convey?
(6) **Visual:** Does this caption mention any visual characteristics of the figure? (Examples include color, shape, direction, size, position, or opacity of any elements in the figure.)

Missed aspects would be highlighted as a warning for users. The aspect detection model is a SciBert [1] model fine-tuned on 3,159 human annotations, which achieved an F1 score of 0.64 [18]. The caption rating and its explanation (Figure 1.F and Figure 1.G) are

[1]Note that in this study, figures classified as "Table" were excluded due to the inherent differences between scientific figures and tables. Despite our belief in SciCapenter's ability to help users write captions for tables, we opted to omit tables (and figures classified as Tables) in this study.

obtained by calling OpenAI's GPT-3.5-turbo API [42] with the prompt shown in Appendix A in a zero-shot manner. The machine-generated captions (Figure 1.H) are generated with the model proposed by Huang et al. [19]. Specifically, we obtained their Pegasus$_{P+O}$ and Pegasus$_{P+O+B}$ model, figure captioning models that generate captions by summarising the figure-mentioning paragraphs. The Pegasus$_{P+O+B}$ model is trained with captions longer than 30 words so it naturally generates longer captions. In SciCapenter, we use the default decoding strategy (beam search with the number of beams = 5) for text generation.

4 USER STUDY

Study Overview and Participant Recruitment. To assess the effectiveness of SciCapenter in assisting users with writing figure captions for academic articles, we recruited 18 participants–3 for a pilot study and 15 for the main study. Many design details of the main study were informed by the pilot study, which we describe in Appendix B.

For recruitment, we leveraged various channels: personal networks, social media posts, Slack group channels, and the university's mailing list. These participants were Ph.D. students from the authors' university, majoring in STEM. Specifically, 6 majored in Computer Science, 7 majored in Informatics, 3 majored in Chemical Engineering and 2 were Mechanical Engineering majors. Ideally, we wanted participants currently writing new papers with figures and captions, but such individuals were scarce. Asking participants to revise and improve poor-quality captions in their published papers also proved challenging, as it required them to admit their papers' weaknesses, complicating recruitment. Consequently, we opted to have participants write figure captions for others' published papers in their fields. Although different from writing captions for figures in their own paper, this method struck a balance between ease of recruitment and task relevance, making it a practical choice for this preliminary study.

Pre-Study Preparation. We asked each participant to provide ten scientific papers from their research domain that they intended to read or had briefly skimmed but had not read in-depth. This approach ensured participants had some contextual understanding of the article but were not biased by the original captions. After receiving these articles, we processed them through SciCapenter, selecting six figures with the lowest quality score rated by SciCapenter. We then manually redacted the captions from these selected papers using Adobe Acrobat Pro (Figure 2). All these materials were prepared before the user study. We tried our best to select only one figure per article. However, `pdfparser2` failed to parse papers from some specific domains much more frequently, so this was not always feasible. In cases where two figures were selected from the same paper for a participant, they received a PDF where both captions were redacted.

Study Procedure and Execution. Most sessions were conducted in-lab, with seven participants joining via Zoom. Each session typically spanned between 1 to 1.5 hours. Participants attending in person used one of the author's laptop, while those joining remotely used their own computers. As SciCapenter is a web-based application, it can be accessed anywhere online. Adobe Acrobat Pro, licensed to all the students in the authors' university, was essential for the study. Hence, we instructed all the remote participants to install it on their computers beforehand.

The steps of the study were as follows: We first introduced participants to the study's objective and procedure, securing their consent. We then offered a brief overview of the SciCapenter and its features. Participants then proceeded through four conditions:

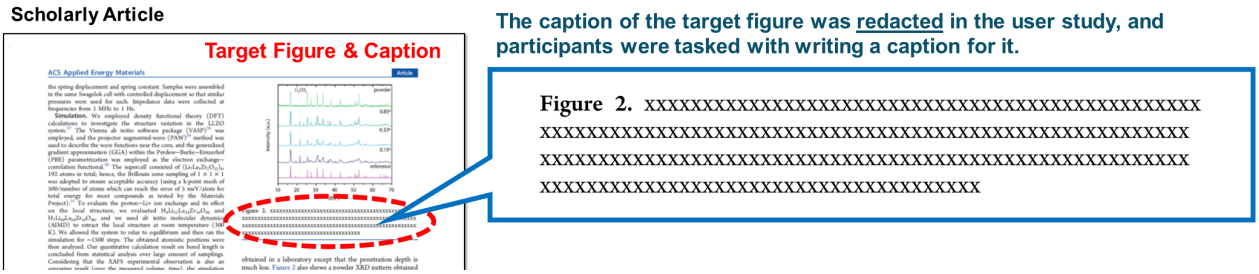
- **Condition 1 (2 figures): Free writing.** Participants were given a PDF with the target caption redacted and asked to draft a caption directly in Adobe Acrobat Pro without any time constraints.
- **Condition 2 (2 figures): Free writing with SciCapenter.** Participants were given a redacted PDF as in Condition 1. However, this time, they wrote captions directly in SciCapenter. Participants were allowed up to two submissions to retrieve SciCapenter's ratings. The submission limit was set to prevent participants from trying to trick the system just to get higher scores.[2]
- **Condition 3 (1 figure):[3] Time-constrained writing.** This condition mirrored Condition 1, but participants were allotted just 8 minutes per figure. The rationale for this time-constrained setup was to test SciCapenter's utility when participants were under pressure for curating captions, such as during last-minute submissions or when captions were given scant attention.
- **Condition 4 (1 figure): Time-constrained writing with SciCapenter.** Same as Condition 2, but participants had only 8 minutes to draft their captions.

Participants sequentially progressed through the conditions: 1 -> 3 -> 2 -> 4. This order, derived from our pilot study (Appendix B), helped prevent confusion and fatigue from switching between Adobe Acrobat Pro and SciCapenter.[4] After each condition, participants completed the NASA-Task Load Index (NASA-TLX) questionnaire [12] via a Google Form. We used a five-point Likert scale (Very Low to Very High) for all scale items, which was used in several prior HCI studies [29, 39]. Upon concluding the whole user study, we conducted a brief, open-ended interview to gather feedback and suggestions from each participant. The duration of the post-interview typically varies, but on average, it lasts about ten minutes. We pose four questions to the participants: **(1) Do you find the system useful, and what aspects do you like or dislike? (2) What is your preferred writing style or approach for figure captions? (3) Do you have any suggestions for our system, and if you could incorporate additional functions, what would they be? (4) Do you have any further feedback?** We

[2]Limiting users to only two attempts for ratings may influence their opinion of the tool and could bias their responses on the NASA task load index. However, our goal was to discourage users from carelessly adjusting their captions without thoughtful consideration. This decision was informed by the writing effort and average caption changes observed in the pilot study.

[3]We initially planned for participants to write captions for two figures under all conditions. However, our pilot study indicated that in time-constrained settings, our setup caused excessive effort and rapid fatigue among users. Therefore, we had participants caption only one figure in time-constrained conditions.

[4]As participants were new to SciCapenter, we understand that this order did not completely remove the learning effect. We believe it makes more sense for participants to first attempt the task without additional support (Condition 1) because writing captions for figures in other people's papers is not a typical writing task.


Scholarly Article
Target Figure & Caption
The caption of the target figure was redacted in the user study, and participants were tasked with writing a caption for it.
Figure 2. xxxx

	Free Writing					Time-Constrained Writing				
	User Only		With System		p-value	User Only		With System		p-value
	Avg.	95% CI	Avg.	95% CI		Avg.	95% CI	Avg.	95% CI	
Mental Demand	2.93	[2.40, 3.47]	2.53	[1.91, 3.16]	0.233	3.80	[3.32, 4.28]	2.53	[1.91, 3.16]	<0.001***
Physical Demand	1.87	[1.40, 2.33]	1.93	[1.40, 2.47]	0.774	2.20	[1.68, 2.72]	1.87	[1.40, 2.33]	0.019
Temporal Demand	1.87	[1.36, 2.37]	1.53	[1.18, 1.89]	0.096	3.20	[2.53, 3.87]	2.40	[1.74, 3.06]	0.005**
Performance	2.47	[1.88, 3.05]	2.47	[1.75, 3.19]	1.000	2.73	[2.12, 3.34]	2.13	[1.48, 2.79]	0.095
Effort	2.87	[2.36, 3.37]	2.40	[1.90, 2.90]	0.150	2.93	[2.40, 3.47]	2.33	[1.84, 2.83]	0.057
Frustration	2.33	[1.62, 3.05]	1.40	[1.12, 1.68]	0.008**	2.73	[2.06, 3.41]	1.67	[1.21, 2.12]	0.001**
Overall	2.39	[2.05, 2.73]	2.04	[1.71, 2.38]	0.046*	2.93	[2.52, 3.34]	2.16	[1.74, 2.57]	<0.001***

Table 1: We leverage the NASA Task Load Index method to assess workload on five-point scales, under conditions of free writing and time-constrained writing, both with and without SciCapenter. A lower score indicates a lower perceived workload. The p-values are obtained by comparing the user-only and with-system settings (paired t-test, two tailed, N=15). The 95% confidence intervals are estimated using t-distribution. With SciCapenter, users had less workload in both free writing and time-constrained writing scenarios, with the most notable improvements seen in the Mental Demand and Frustration.

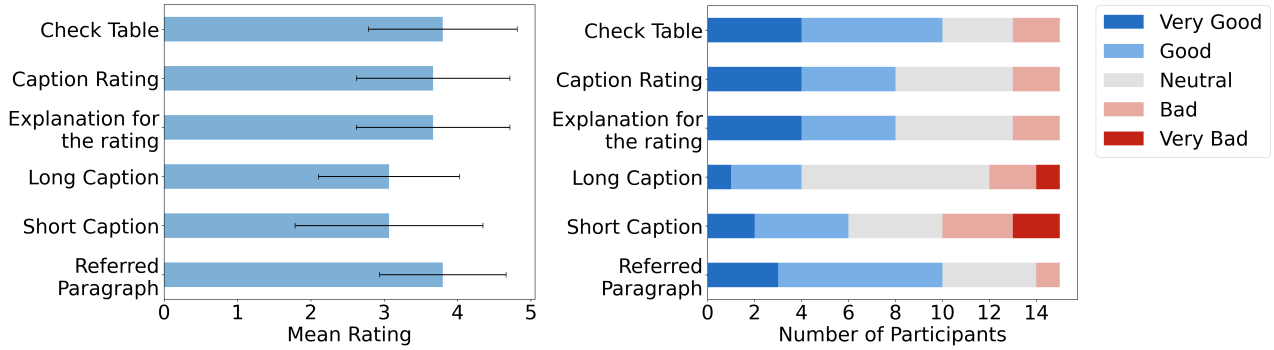


Figure 3: Comparison between six different elements provided by SciCapenter. Left figure shows the mean rating and standard deviation of a five-point scale for different elements. The check table and referred paragraph were rated highest, while short and long caption had the lowest score, indicating they were the least favored elements according to the participants. Right figure shows a breakdown of the five-point scale with different colors representing each rating. The short caption exhibit a more varied distribution of opinions.

by themselves. (P10)". Some suggested SciCapenter to provide example captions for similar figures: "... you can generate a typical caption from other papers that have already been published. (P7)", "I would be curious about how other people do for similar figures. ..., I would like to see what people mention in similar domains of papers for similar formats of figures. (P8)" Some desired a template that they could simply fill in information: "..., you have standard (templates), and you just need to fill in the details. (P7)", "..., you want some, like standard structure. Just a structure, not the exact paragraphs that you need to mention, ... (P11)".

6 DISCUSSION

Design Recommendations. Our research indicates that while machine-generated texts may not always match the quality of human-written content, they might still offer assistance in writing tasks. To address the potential shortcomings of machine-generated texts, SciCapenter incorporated an abundance of contextual information, such as caption traits, caption ratings, explanations for these ratings, and references to relevant figures, to aid users. Based on our study results and user interviews, we offer two design recommendations for writing assistants targeting figure captions:

- **Present contextual or analytical information relevant to the writing task.** While machine-generated captions were generally deemed useful, our results suggested that the benefits of presenting contextual (*e.g.*, figure-mentioning paragraphs) or analytical (*e.g.*, check tables, ratings) information relevant to the writing task might be more prevalent (Figure 3).
- **Provide concrete, actionable suggestions for writing.** SciCapenter analyzed the current situation without offering specific guidance on enhancing a caption or providing fillable caption templates, leading to many participant complaints (Section 5). Future developers of writing assistants for caption writing should aim to provide more targeted, actionable suggestions. Continued research and development are crucial for this.

Comparison of Caption Quality. The study was conducted before we had access to GPT-4V [43], one of the strongest Large Vision-Language Models (LVLM), raising an interesting question: **Could**

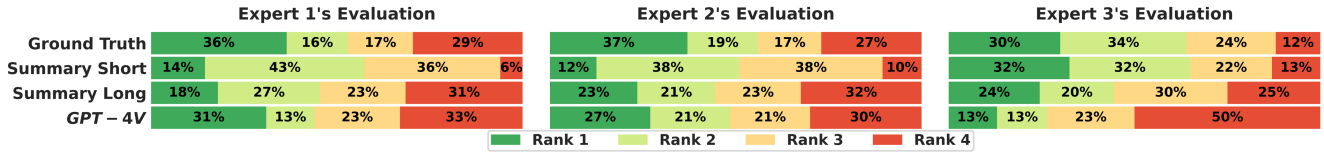


Figure 4: Comparative evaluation of caption quality by three experts, where each caption type—Ground Truth, Summary Short, Summary Long, and GPT-4V—is rated on a scale from rank 1 (highest) to rank 4 (lowest). Both Expert 1 and Expert 2 rated Ground Truth caption as rank 1 most frequently, while Expert 3 had a preference for Summary Short. Notably, Expert 3 rated GPT-4V the lowest, rarely giving it a rank 1, whereas both Expert 1 and Expert 2 often considered GPT-4V as their second choice for rank 1. The variations in evaluations reflect differing perspectives on caption quality and suggest that while Ground Truth captions are generally preferred, there's a significant disparity in how each expert rates the machine-generated captions.

GPT-4V produce significantly better captions than the models used in SciCapenter, potentially altering our study's conclusions? To explore this, we recruited three professional academic paper editors to assess the quality of captions generated by various models.

From the SciCap Challenge dataset [17], we randomly selected 200 figure-caption pairs. For each figure, we generated two captions using models from SciCapenter (Summary Short and Summary Long), one caption using GPT-4V by prompting it with the figure image and figure-mentioning paragraphs, and we also included the original author-written caption (Ground Truth). To ensure a fair comparison in our study, GPT-4V was prompted to generate captions no longer than the author-written captions, acknowledging that readers often favor longer captions. This length constraint aimed to minimize the influence of caption length on quality assessment.

We recruited three professional editors (Expert 1, 2, and 3) through UpWork,[5] all specializing in technical academic articles and native American English speakers. Their backgrounds include one with over ten years of editing experience and a Ph.D. in Comparative Literature, and two from the STEM fields—one in Theoretical Astrophysics and another in Neuroscience—with extensive experience in editing, proofreading, and publishing academic papers. They ranked the captions based on their effectiveness in clarifying the figure's intended message. The results are shown in Figure 4. Under the caption length constraints, GPT-4V did not outperform SciCapenter's models drastically. Although Experts 1 and 2 preferred GPT-4V's generation over that of SciCapenter's models, Expert 3 did not favor it.

Limitations. Our work demonstrates that AI technologies can support caption writing, yet we acknowledge the following limitations:

(1) Our study involved participants writing figure captions for other people's published papers rather than crafting captions for their own work. This scenario diverged from typical use cases of figure caption writing support systems. The study was designed this way to accommodate the difficulty in recruiting individuals who are actively writing new scholarly articles.
(2) Despite efforts to mitigate learning effects by altering the condition order based on the pilot study (Appendix B), our study procedure could still introduce potential biases. For instance, participants had to learn the uncommon task of writing captions for other people's papers in the first condition they encountered. Furthermore, all participants first wrote the captions without SciCapenter and then used SciCapenter, which may have amplified the system's benefits.
(3) Our study only used NASA-TLX. Future research can extend the measures to not only NASA-TLX but to other measures, such as a sense of authority or confidence level, to understand the benefit of SciCapenter in multiple aspects.
(4) As a preliminary study and early prototype, the lack of consideration for specific items from the Check Table in both generated captions and explanations for ratings suggests a need for better integration with the tool's features.
(5) Our study did not assess the quality of the resulting captions, leaving unanswered whether SciCapenter improved caption quality. This was due to the high cost and need for domain experts to evaluate caption usefulness. Further studies are needed to determine writing assistants' impact on caption quality.

[5]UpWork: https://www.upwork.com/

7 CONCLUSION AND FUTURE WORK

In this paper, we present SciCapenter, a system designed to assist authors in crafting captions for scientific figures within scholarly articles. Given a scholarly article, SciCapenter identifies and extracts figures, isolates paragraphs that mention these figures, generates captions based on these paragraphs, and then rates the quality of these captions. Our user study showed that SciCapenter reduced users' cognitive load while composing figure captions. Considering this is an initial effort in aiding caption creation, our future goal is to develop a more accessible version of SciCapenter, potentially in the form of a web browser plugin. We aim to conduct a broader deployment study to understand how researchers might integrate this tool into their academic writing.

REFERENCES

[1] Iz Beltagy, Kyle Lo, and Arman Cohan. 2019. SciBERT: A pretrained language model for scientific text. *arXiv preprint arXiv:1903.10676* (2019).
[2] John Bransford. 1979. Human cognition: Learning, understanding, and remembering. *(No Title)* (1979).
[3] Yu-Ying Chang and John M Swales. 2014. Informal elements in English academic writing: threats or opportunities for advanced non-native speakers? In *Writing: Texts, processes and practices*. Routledge, 145–167.
[4] Charles Chen, Ruiyi Zhang, Eunyee Koh, Sungchul Kim, Scott Cohen, and Ryan Rossi. 2020. Figure captioning with relation maps for reasoning. In *Proceedings of*

the IEEE/CVF Winter Conference on Applications of Computer Vision. 1537–1545.

[5] Jiwon Choi and Jaemin Jo. 2022. Intentable: A mixed-initiative system for intent-based chart captioning. In *2022 IEEE Visualization and Visual Analytics (VIS)*. IEEE, 40–44.

[6] Jinho Choi, Sanghun Jung, Deok Gun Park, Jaegul Choo, and Niklas Elmqvist. 2019. Visualizing for the non-visual: Enabling the visually impaired to use visualization. In *Computer Graphics Forum*, Vol. 38. Wiley Online Library, 249–260.

[7] Christopher Clark and Santosh Divvala. 2016. Pdffigures 2.0: Mining figures from research papers. In *Proceedings of the 16th ACM/IEEE-CS on Joint Conference on Digital Libraries*. 143–152.

[8] Susan M Conrad. 1996. Investigating academic texts with corpus-based techniques: An example from biology. *Linguistics and education* 8, 3 (1996), 299–326.

[9] Ignacio Garcia and María Isabel Pena. 2011. Machine translation-assisted language learning: writing for beginners. *Computer Assisted Language Learning* 24, 5 (2011), 471–487.

[10] Yashesh Gaur, Walter S Lasecki, Florian Metze, and Jeffrey P Bigham. 2016. The effects of automatic speech recognition quality on human transcription latency. In *Proceedings of the 13th International Web for All Conference*. 1–8.

[11] Li Gong, Josep Crego, and Jean Senellart. 2019. Enhanced Transformer Model for Data-to-Text Generation. In *Proceedings of the 3rd Workshop on Neural Generation and Translation*, Alexandra Birch, Andrew Finch, Hiroaki Hayashi, Ioannis Konstas, Thang Luong, Graham Neubig, Yusuke Oda, and Katsuhito Sudoh (Eds.). Association for Computational Linguistics, Hong Kong, 148–156. https://doi.org/10.18653/v1/D19-5615

[12] Sandra G Hart. 2006. NASA-task load index (NASA-TLX); 20 years later. In *Proceedings of the human factors and ergonomics society annual meeting*, Vol. 50. Sage publications Sage CA: Los Angeles, CA, 904–908.

[13] Mary Hegarty and Marcel-Adam Just. 1993. Constructing mental models of machines from text and diagrams. *Journal of memory and language* 32, 6 (1993), 717–742.

[14] John Hinds, U Connor, and RB Kaplan. 1987. Reader versus writer responsibility: A new typology. *Landmark essays on ESL writing* (1987), 63–74.

[15] Sameera Horawalavithana, Sai Munikoti, Ian Stewart, and Henry Kvinge. 2023. Scitune: Aligning large language models with scientific multimodal instructions. *arXiv preprint arXiv:2307.01139* (2023).

[16] Ting-Yao Hsu, C Lee Giles, and Ting-Hao Huang. 2021. SciCap: Generating Captions for Scientific Figures. In *Findings of the Association for Computational Linguistics: EMNLP 2021*. Association for Computational Linguistics, Punta Cana, Dominican Republic, 3258–3264. https://doi.org/10.18653/v1/2021.findings-emnlp.277

[17] Ting-Yao Hsu, Yi-Li Hsu, Shaurya Rohatgi, Ryan Rossi, Sungchul Kim, Ani Nenkova, Lun-Wei Ku, Huijuan Xu, C. Giles, and Ting-Hao Huang. 2023. The 1st Scientific Figure Captioning (SciCap) Challenge. http://scicap.ai/.

[18] Ting-Yao Hsu, Chieh-Yang Huang, Ryan Rossi, Sungchul Kim, C. Giles, and Ting-Hao Huang. 2023. GPT-4 as an Effective Zero-Shot Evaluator for Scientific Figure Captions. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, Houda Bouamor, Juan Pino, and Kalika Bali (Eds.). Association for Computational Linguistics, Singapore, 5464–5474. https://doi.org/10.18653/v1/2023.findings-emnlp.363

[19] Chieh-Yang Huang, Ting-Yao Hsu, Ryan Rossi, Ani Nenkova, Sungchul Kim, Gromit Yeuk-Yin Chan, Eunyee Koh, Clyde Lee Giles, and Ting-Hao'Kenneth' Huang. 2023. Summaries as Captions: Generating Figure Captions for Scientific Documents with Automated Text Summarization. *arXiv preprint arXiv:2302.12324* (2023).

[20] KV Jobin, Ajoy Mondal, and CV Jawahar. 2019. Docfigure: A dataset for scientific document figure classification. In *2019 International Conference on Document Analysis and Recognition Workshops (ICDARW)*, Vol. 1. IEEE, 74–79.

[21] Kushal Kafle, Brian Price, Scott Cohen, and Christopher Kanan. 2018. Dvqa: Understanding data visualizations via question answering. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 5648–5656.

[22] Samira Ebrahimi Kahou, Vincent Michalski, Adam Atkinson, Ákos Kádár, Adam Trischler, and Yoshua Bengio. 2017. Figureqa: An annotated figure dataset for visual reasoning. *arXiv preprint arXiv:1710.07300* (2017).

[23] Shankar Kantharaj, Xuan Long Do, Rixie Tiffany Leong, Jia Qing Tan, Enamul Hoque, and Shafiq Joty. 2022. OpenCQA: Open-ended Question Answering with Charts. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, Yoav Goldberg, Zornitsa Kozareva, and Yue Zhang (Eds.). Association for Computational Linguistics, Abu Dhabi, United Arab Emirates, 11817–11837. https://doi.org/10.18653/v1/2022.emnlp-main.811

[24] Shankar Kantharaj, Rixie Tiffany Leong, Xiang Lin, Ahmed Masry, Megh Thakkar, Enamul Hoque, and Shafiq Joty. 2022. Chart-to-Text: A Large-Scale Benchmark for Chart Summarization. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Smaranda Muresan, Preslav Nakov, and Aline Villavicencio (Eds.). Association for Computational Linguistics, Dublin, Ireland, 4005–4023. https://doi.org/10.18653/v1/2022.acl-long.277

[25] Zeba Karishma, Shaurya Rohatgi, Kavya Shrinivas Puranik, Jian Wu, and C Lee Giles. 2023. ACL-Fig: A Dataset for Scientific Figure Classification. *arXiv preprint arXiv:2301.12293* (2023).

[26] Dae Hyun Kim, Vidya Setlur, and Maneesh Agrawala. 2021. Towards Understanding How Readers Integrate Charts and Captions: A Case Study with Line Charts. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*.

[27] Dae Hyun Kim, Vidya Setlur, and Maneesh Agrawala. 2021. Towards understanding how readers integrate charts and captions: A case study with line charts. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–11.

[28] Philipp Koehn and Barry Haddow. 2009. Interactive assistance to human translators using statistical machine translation methods. In *Proceedings of Machine Translation Summit XII: Papers*.

[29] Vivian Lai, Samuel Carton, Rajat Bhatnagar, Q Vera Liao, Yunfeng Zhang, and Chenhao Tan. 2022. Human-ai collaboration via conditional delegation: A case study of content moderation. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–18.

[30] Andrew Large, Jamshid Beheshti, Alain Breuleux, and Andre Renaud. 1995. Multimedia and comprehension: The relationship among text, animation, and captions. *Journal of the American society for information science* 46, 5 (1995), 340–347.

[31] Kenton Lee, Mandar Joshi, Iulia Raluca Turc, Hexiang Hu, Fangyu Liu, Julian Martin Eisenschlos, Urvashi Khandelwal, Peter Shaw, Ming-Wei Chang, and Kristina Toutanova. 2023. Pix2struct: Screenshot parsing as pretraining for visual language understanding. In *International Conference on Machine Learning*. PMLR, 18893–18912.

[32] Shengzhi Li and Nima Tajbakhsh. 2023. Scigraphqa: A large-scale synthetic multi-turn question-answering dataset for scientific graphs. *arXiv preprint arXiv:2308.03349* (2023).

[33] Yanna Lin, Haotian Li, Leni Yang, Aoyu Wu, and Huamin Qu. 2023. Inksight: Leveraging sketch interaction for documenting chart findings in computational notebooks. *IEEE Transactions on Visualization and Computer Graphics* (2023).

[34] Can Liu, Yuhan Guo, and Xiaoru Yuan. 2023. Autotitle: An interactive title generator for visualizations. *IEEE Transactions on Visualization and Computer Graphics* (2023).

[35] Fangyu Liu, Francesco Piccinno, Syrine Krichene, Chenxi Pang, Kenton Lee, Mandar Joshi, Yasemin Altun, Nigel Collier, and Julian Martin Eisenschlos. 2022. Matcha: Enhancing visual language pretraining with math reasoning and chart derendering. *arXiv preprint arXiv:2212.09662* (2022).

[36] Anita Mahinpei, Zona Kostic, and Chris Tanner. 2022. Linecap: Line charts for data visualization captioning models. In *2022 IEEE Visualization and Visual Analytics (VIS)*. IEEE, 35–39.

[37] Ahmed Masry, Parsa Kavehzadeh, Xuan Long Do, Enamul Hoque, and Shafiq Joty. 2023. UniChart: A Universal Vision-language Pretrained Model for Chart Comprehension and Reasoning. *arXiv preprint arXiv:2305.14761* (2023).

[38] Ahmed Masry, Do Xuan Long, Jia Qing Tan, Shafiq Joty, and Enamul Hoque. 2022. ChartQA: A benchmark for question answering about charts with visual and logical reasoning. *arXiv preprint arXiv:2203.10244* (2022).

[39] Florian Mathis, John H Williamson, Kami Vaniea, and Mohamed Khamis. 2021. Fast and secure authentication in virtual reality using coordinated 3d manipulation and pointing. *ACM Transactions on Computer-Human Interaction (ToCHI)* 28, 1 (2021), 1–44.

[40] Gwen C Nugent. 1983. Deaf students' learning from captioned instruction: The relationship between the visual and caption display. *The Journal of Special Education* 17, 2 (1983), 227–234.

[41] Jason Obeid and Enamul Hoque. 2020. Chart-to-Text: Generating Natural Language Descriptions for Charts by Adapting the Transformer Model. In *Proceedings of the 13th International Conference on Natural Language Generation*, Brian Davis, Yvette Graham, John Kelleher, and Yaji Sripada (Eds.). Association for Computational Linguistics, Dublin, Ireland, 138–147. https://doi.org/10.18653/v1/2020.inlg-1.20

[42] OpenAI. 2022. GPT-3.5: Language Models are Few-Shot Learners. https://platform.openai.com/docs/models/gpt-3-5.

[43] OpenAI. 2023. GPT-4V(ision) System Card. https://api.semanticscholar.org/CorpusID:263218031

[44] William D Page. 1974. The author and the reader in writing and reading. *Research in the Teaching of English* 8, 2 (1974), 170–183.

[45] Xin Qian, Eunyee Koh, Fan Du, Sungchul Kim, Joel Chan, Ryan A Rossi, Sana Malik, and Tak Yeon Lee. 2021. Generating accurate caption units for figure captioning. In *Proceedings of the Web Conference 2021*. 2792–2804.

[46] Noah Siegel, Zachary Horvitz, Roie Levin, Santosh Divvala, and Ali Farhadi. 2016. Figureseer: Parsing result-figures in research papers. In *Computer Vision–ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11–14, 2016, Proceedings, Part VII 14*. Springer, 664–680.

[47] Ashish Singh, Prateek Agarwal, Zixuan Huang, Arpita Singh, Tong Yu, Sungchul Kim, Victor Bursztyn, Nikos Vlassis, and Ryan A Rossi. 2023. FigCaps-HF: A Figure-to-Caption Generative Framework and Benchmark with Human Feedback. *arXiv preprint arXiv:2307.10867* (2023).

[48] Chase Stokes, Vidya Setlur, Bridget Cogley, Arvind Satyanarayan, and Marti A Hearst. 2022. Striking a balance: reader takeaways and preferences when integrating text and charts. *IEEE Transactions on Visualization and Computer Graphics* 29, 1 (2022), 1233–1243.
[49] Benny J Tang, Angie Boggust, and Arvind Satyanarayan. 2023. Vistext: A benchmark for semantically rich chart captioning. *arXiv preprint arXiv:2307.05356* (2023).
[50] Yanan Wang and Yea-Seul Kim. 2023. Making data-driven articles more accessible: An active preference learning approach to data fact personalization. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference*. 1353–1366.
[51] Zhishen Yang, Raj Dabre, Hideki Tanaka, and Naoaki Okazaki. 2023. SciCap+: A Knowledge Augmented Dataset to Study the Challenges of Scientific Figure Captioning. *arXiv preprint arXiv:2306.03491* (2023).
[52] Jiabo Ye, Anwen Hu, Haiyang Xu, Qinghao Ye, Ming Yan, Yuhao Dan, Chenlin Zhao, Guohai Xu, Chenliang Li, Junfeng Tian, et al. 2023. mplug-docowl: Modularized multimodal large language model for document understanding. *arXiv preprint arXiv:2307.02499* (2023).

A PROMPTS USED

The following is the prompt used by SciCapenter to generate the caption rating and its explanation (Figure 1.F and Figure 1.G):

```
Given the paragraph and caption below, please
rate the level of usefulness of the caption
from 1 to 6 based on how well the caption
could help readers understand the important
information. 6 is the highest; 1 is the
lowest. Please also explain your rating.
Paragraph: "[paragraph]"
Caption: "[caption]"
```

where `[paragraph]` and `[caption]` are the placeholder for the referred paragraph and written caption, respectively. Note that we choose GPT-3.5-turbo over GPT-4 due to its higher stability and shorter response time.

B PILOT STUDY AND CHANGES MADE

The first three study trials were treated as the pilot study. Two of the participants majored in Informatics, and one in Computer Science. The pilot study yielded several valuable insights that informed adjustments made in our main study:

- In the pilot study, the task sequence was ordered 2 -> 4 -> 1 -> 3, with the SciCapenter conditions proceeding first. We observed that after participants viewed the six aspects reported in the check table (Figure 1.E), their writings were influenced, for example, to include takeaway messages and figures' visual features. To minimize this influence, we changed the task order by asking participants to proceed with the conditions without SciCapenter first in the formal study. We also included extra instructions clarifying that the six aspects are suggestions, not requirements.
- In the pilot study, the rating's explanation (Figure 1.G) was only visible when users hovered over a small button. We found that participants frequently ignored this explanation. As a result, we modified the interface to display the explanation directly.
- The missing aspects in the check table (Figure 1.E) were originally tagged with the icon "X". Users expressed concerns about such a strong rejection symbol. We, therefore, replaced it with a triangle alert symbol.
- The time constraint for Condition 3 and Condition 4 in the pilot study was set to five minutes. We found that participants were quite stressed and could barely finish the caption. So, we extended the time limit to 8 minutes.
- GPT-4 was used for automatic caption evaluation (Figure 1.F and Figure 1.G) in the pilot study. However, the API's response time was quite high. Thus, we changed to GPT-3.5-turbo in the main study.